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Studierichting: Informatica
Oefeningenreeks 4A nr. 39.3

$$\int \frac{2dx}{(x^2+2)^3}$$

$$x = \sqrt{2}z$$

$$= 2\int \frac{1}{(2(z^2+1))^3} \cdot \sqrt{2}dz$$

$$= 2\int \frac{1}{8(z^2+1)^3} \cdot \sqrt{2}dz$$

$$= 2 \cdot \frac{1}{8} \cdot \sqrt{2} \int \frac{dz}{(z^2+1)^3}$$

$$= \frac{\sqrt{2}}{4} \int \frac{dz}{(z^2+1)^3}$$

$$= \frac{\sqrt{2}}{4} M_3(z)$$

$$M_2 = \frac{1}{2}M_1 + \frac{z}{2(z^2+1)} = \frac{1}{2}\operatorname{Bgtan}(z) + \frac{z}{2(z^2+1)} + C$$

$$M_3 = \frac{3}{4}M_2 + \frac{z}{4(z^2+1)^2} = \frac{3}{4}\left(\frac{1}{2}\operatorname{Bgtan}(z) + \frac{z}{2(z^2+1)}\right) + \frac{z}{4(z^2+1)^2} + C$$

$$= \frac{3}{8}\operatorname{Bgtan}(z) + \frac{3z}{8(z^2+1)} + \frac{z}{4(z^2+1)^2} + C$$

$$= \frac{\sqrt{2}}{4}\left(\frac{3}{8}\operatorname{Bgtan}(z) + \frac{3z}{8(z^2+1)} + \frac{z}{4(z^2+1)^2}\right) + C$$

$$= \frac{\sqrt{2}}{4}\left(\frac{3}{8}\operatorname{Bgtan}\left(\frac{x}{\sqrt{2}}\right) + \frac{3 \cdot \frac{x}{\sqrt{2}}}{8 \cdot \frac{x^2+2}{2}} + \frac{\frac{x}{\sqrt{2}}}{4 \cdot \frac{(x^2+2)^2}{4}}\right) + C$$

$$= \frac{\sqrt{2}}{4}\left(\frac{3}{8}\operatorname{Bgtan}\left(\frac{x}{\sqrt{2}}\right) + \frac{3x}{4\sqrt{2}(x^2+2)} + \frac{x}{\sqrt{2}(x^2+2)^2}\right) + C$$

$$= \frac{3\sqrt{2}}{32}\operatorname{Bgtan}\left(\frac{x}{\sqrt{2}}\right) + \frac{3x}{16(x^2+2)} + \frac{x}{4(x^2+2)^2} + C$$

References